ECON 0150 | Economic Data Analysis

The economist's data analysis skillset.

Part 2 | Wrap Up

The Part 2 Toolkit

Three figures, each matched to a set of variables

Part	Variables	Figure	Later becomes…
2.1	Numerical by Numerical	Scatterplot	Regression (Part 4)
2.2	Numerical by Categorical	Boxplot (+ stripplot)	Two-sample test (Part 3)
2.3	Numerical by Numerical by Categorical	Scatter colored by category	Regression with a control (Part 5)

Before we draw, we prepare: **filter** rows that don't belong, **log** skewed variables, **merge** data from different sources.

Choose the Visualization

Exercise 1: Commuting and Gas

What are the variable types, and what figure would you draw?

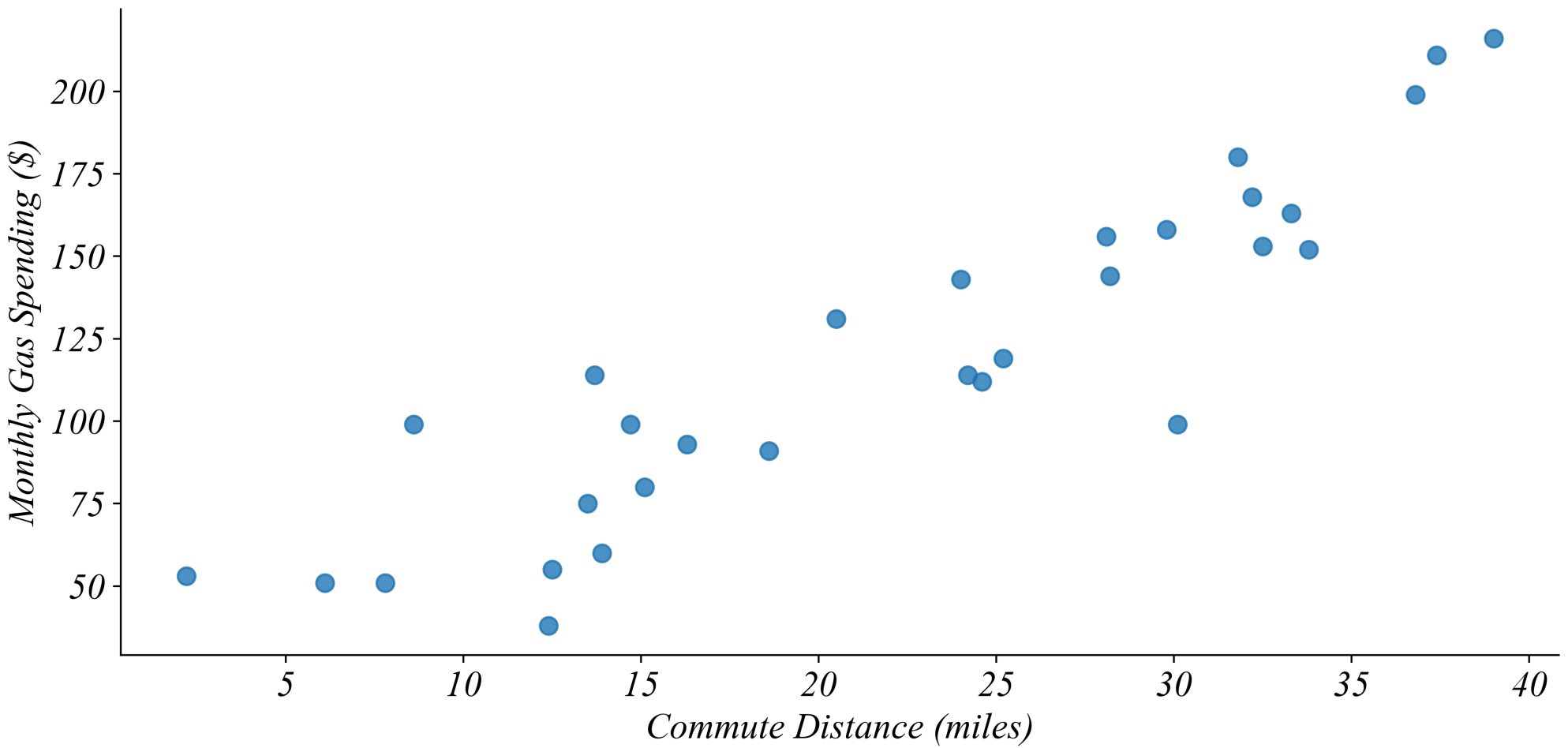
A dataset on commute distance and monthly gas spending for 30 workers.

Commute_Miles	Gas_Spend
2.2	53
37.4	211
31.8	180
14.7	99
28.1	156

Variable Types: Numerical by Numerical

Exercise 1: Commuting and Gas

What are the variable types, and what figure would you draw?



A ***Scatterplot*** shows how one numerical variable moves with another.

Exercise 2: Rent by Area

What are the variable types, and what figure would you draw?

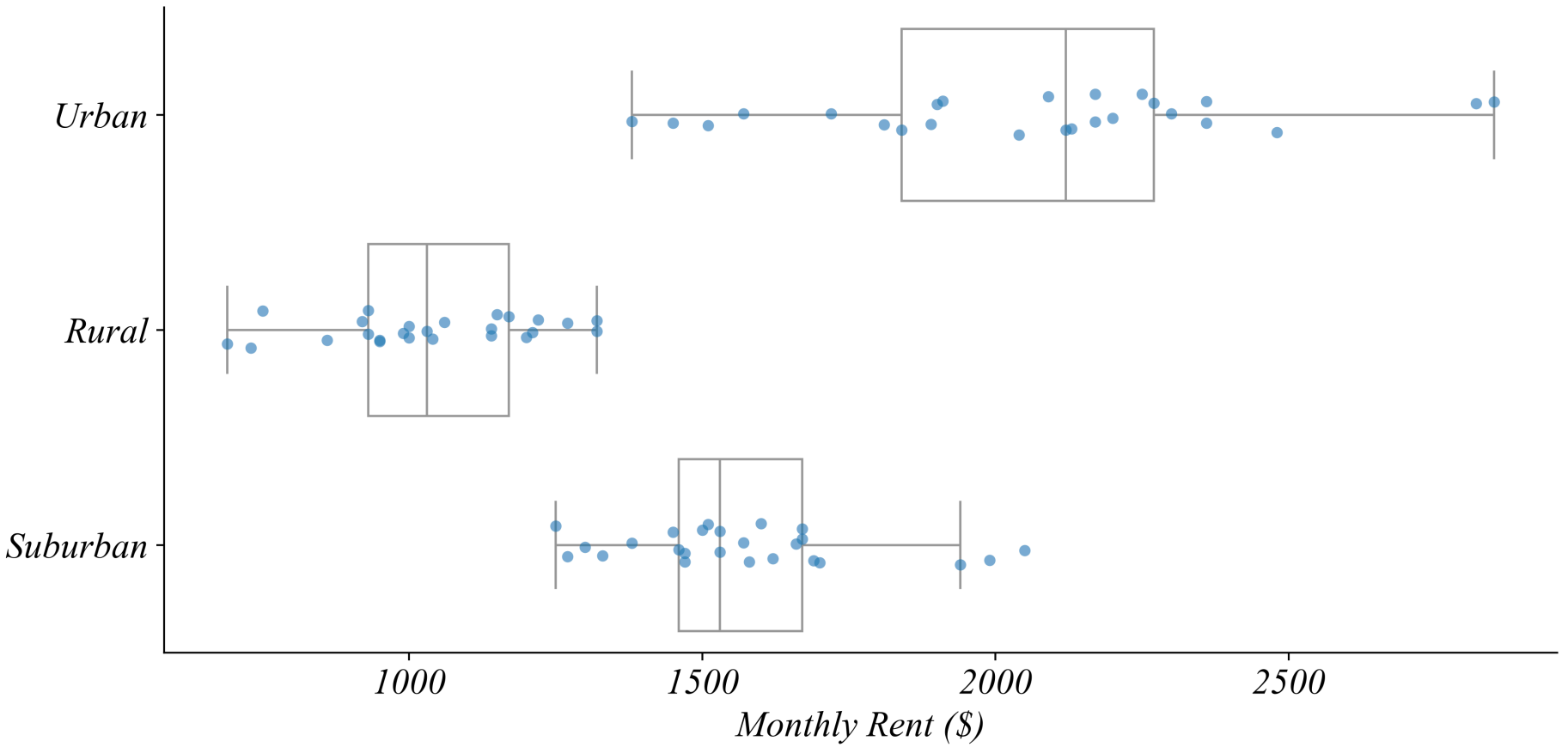
A dataset on monthly rent for 75 apartments.

Area	Rent
Urban	2130
Rural	1060
Urban	1720
Rural	860
Rural	950

Variable Types: Numerical by Categorical

Exercise 2: Rent by Area

What are the variable types, and what figure would you draw?



A ***Boxplot by category*** compares the distribution across groups. The ***stripplot*** shows each apartment.

Exercise 3: Studying and Exam Scores

What are the variable types, and what figure would you draw?

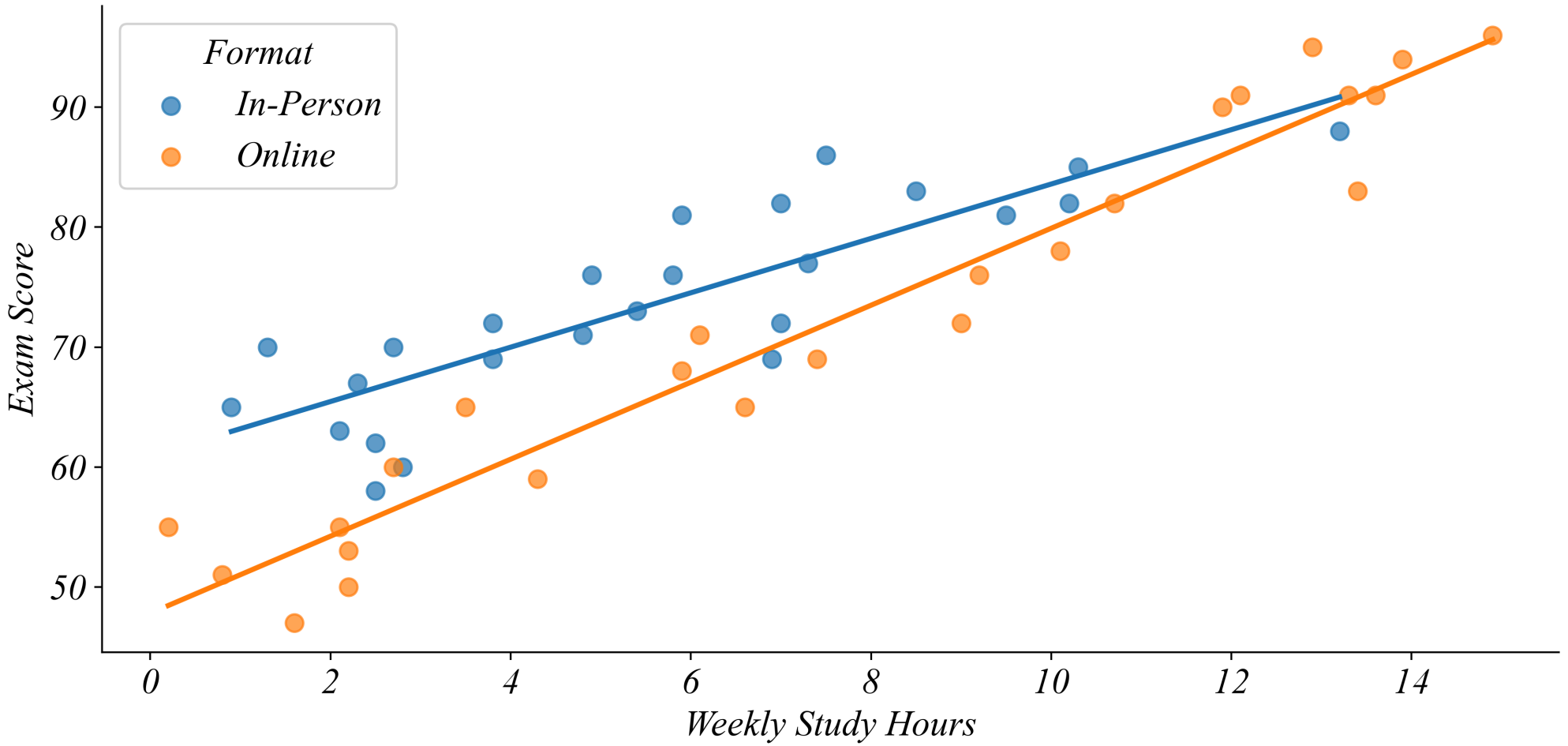
A dataset on weekly study hours and exam scores for students in two course formats.

Format	Study_Hours	Exam_Score
Online	3.5	65
Online	13.4	83
Online	0.8	51
In-Person	0.9	65
In-Person	1.3	70

Variable Types: Numerical by Numerical by Categorical

Exercise 3: Studying and Exam Scores

What are the variable types, and what figure would you draw?



A ***Scatter colored by category*** shows whether the *relationship* differs between groups.

Exercise 4: Inflation and Unemployment

What are the variable types, and what figure would you draw?

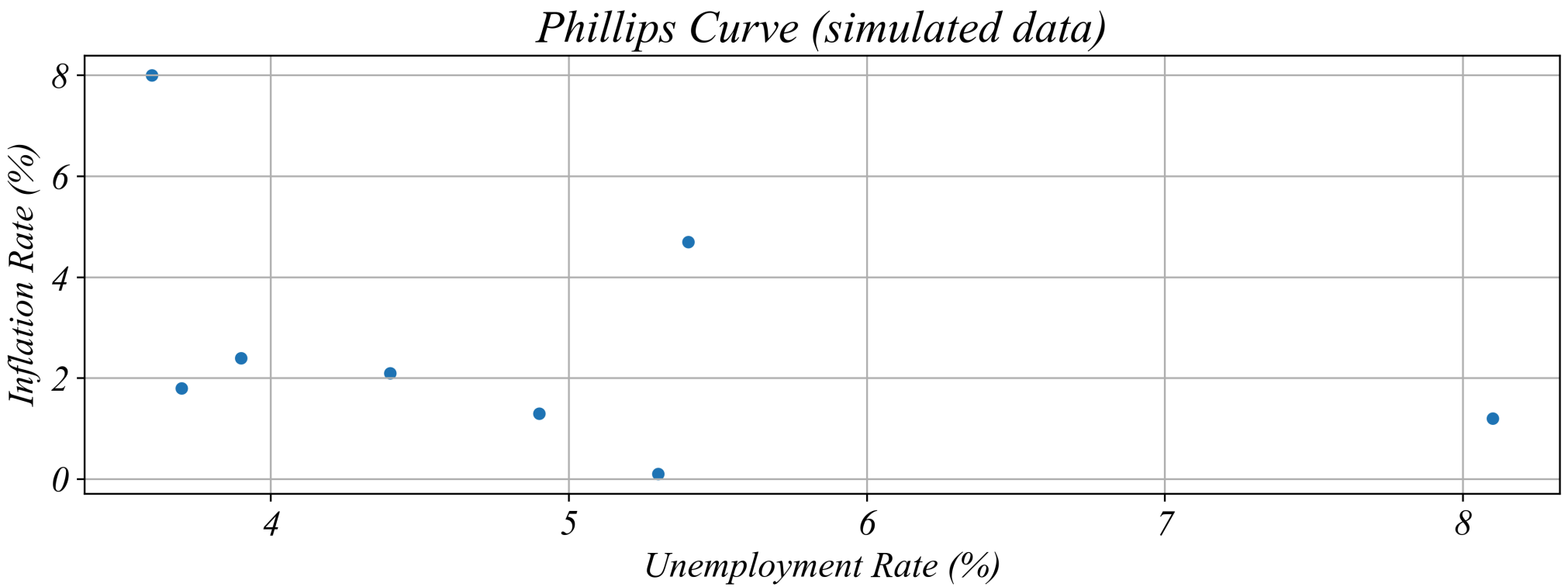
US unemployment and inflation rates each year since 2015.

year	unemployment	inflation
2015	5.3	0.1
2016	4.9	1.3
2017	4.4	2.1
2018	3.9	2.4
2019	3.7	1.8

Variable Types: Numerical by Numerical, measured over time

Exercise 4: Inflation and Unemployment

What are the variable types, and what figure would you draw?

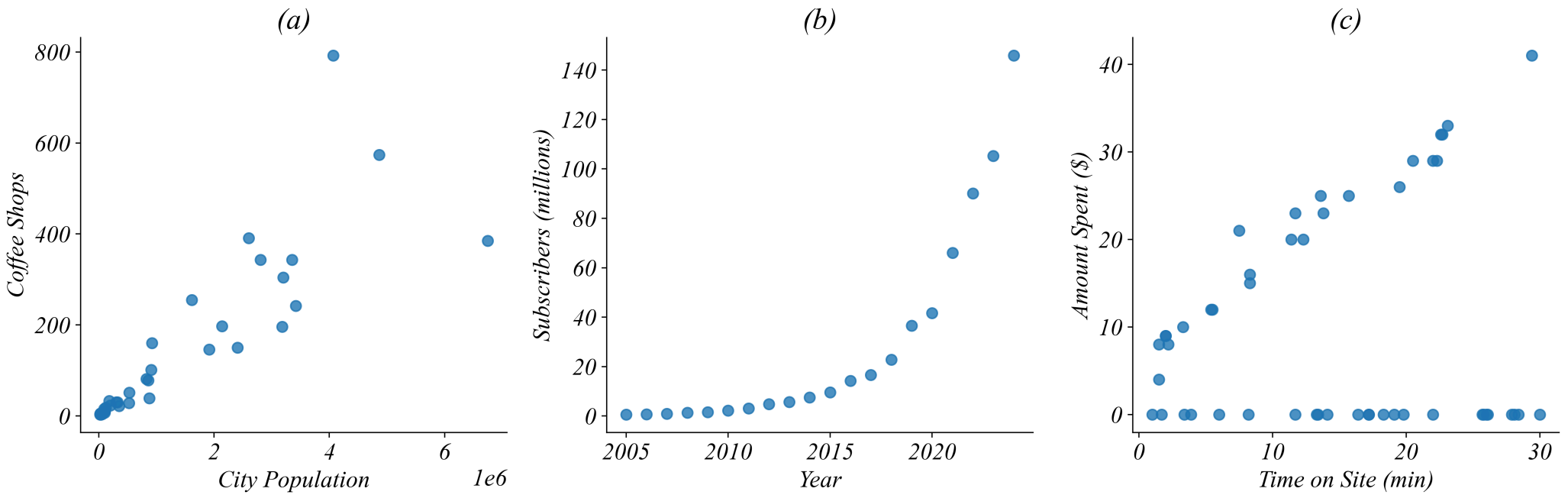


A ***Scatterplot*** works for timeseries too, like coffee and oil prices in Part 2.1. Labeling the points with years keeps the time order.

Fix the Figure

Fix the Figure

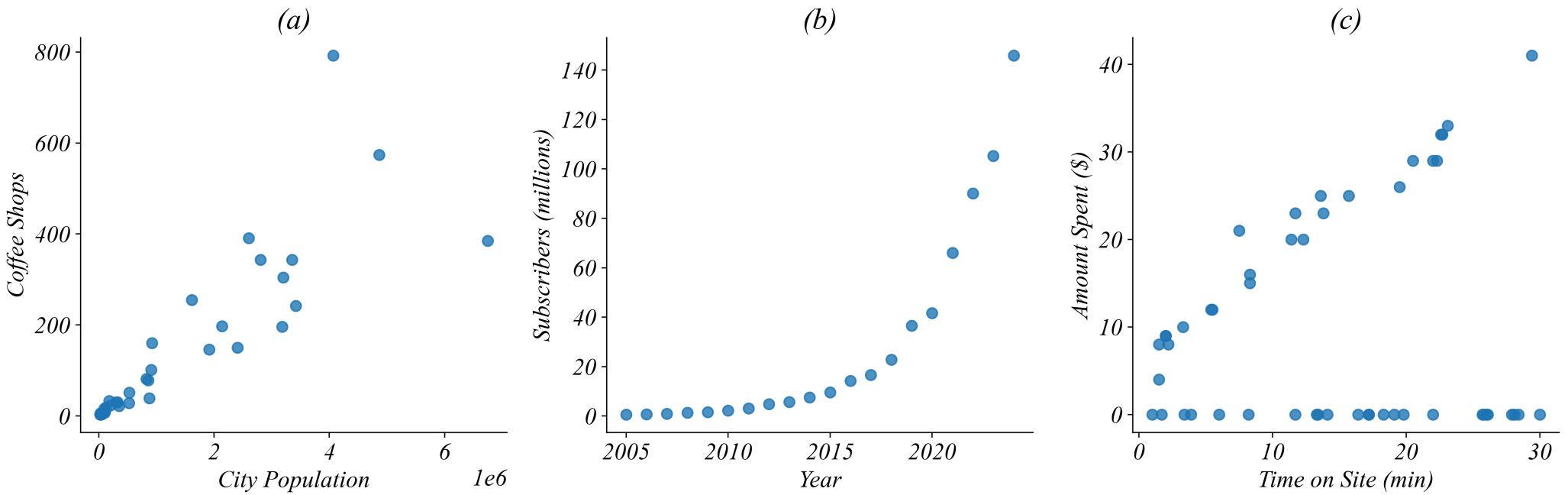
Improve the figure for the question.



- ***(a)*** *How does the number of coffee shops change with city size, for small cities as well as large ones?*
- ***(b)*** *Do subscribers grow at a steady rate, or does the growth rate change over time?*
- ***(c)*** *Among customers who buy something, do longer visits lead to bigger purchases?*

Fix the Figure

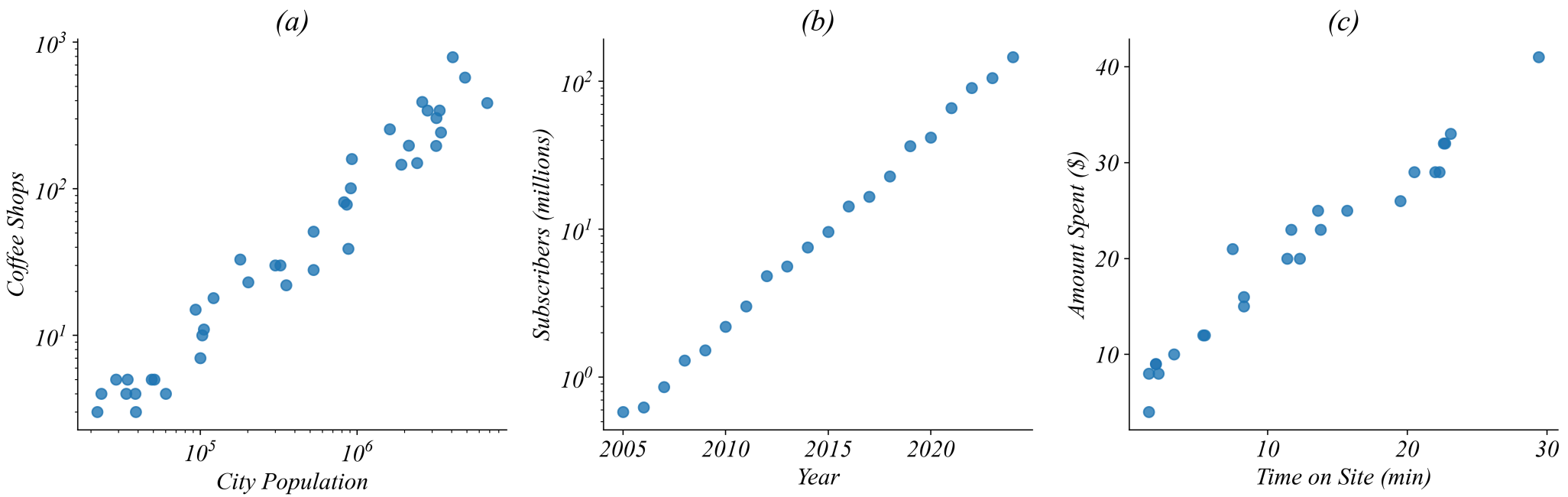
What does each figure show well, and what does it make hard to see?



- ***(a)*** *Most cities clump in the bottom-left corner. We can't see the small cities at all.*
- ***(b)*** *The curve is accurate: growth is speeding up. But it's hard to judge by eye whether a curve bends at a steady rate.*
- ***(c)*** *The zeros are real and useful: most visitors buy nothing. But our question is only about the buyers.*

Fix the Figure

Log both axes, log one axis, or filter



- ***(a)*** *Log* ***both*** *axes: small and large cities spread out, and each step is a multiple of the last.*
- ***(b)*** *Log the* ***y-axis only****: a steady percentage growth rate becomes a straight line.*
- ***(c) Filter*** *to buyers: among those who spend, longer visits go with bigger purchases.*

Log Scales

Reading Log Scales

Equal distances on a log scale mean equal multiples

a. On a log10 scale, each one-unit increase means the original value has been…

multiplied by 10

b. On a log2 scale, moving up 3 units means the original value has been multiplied by…

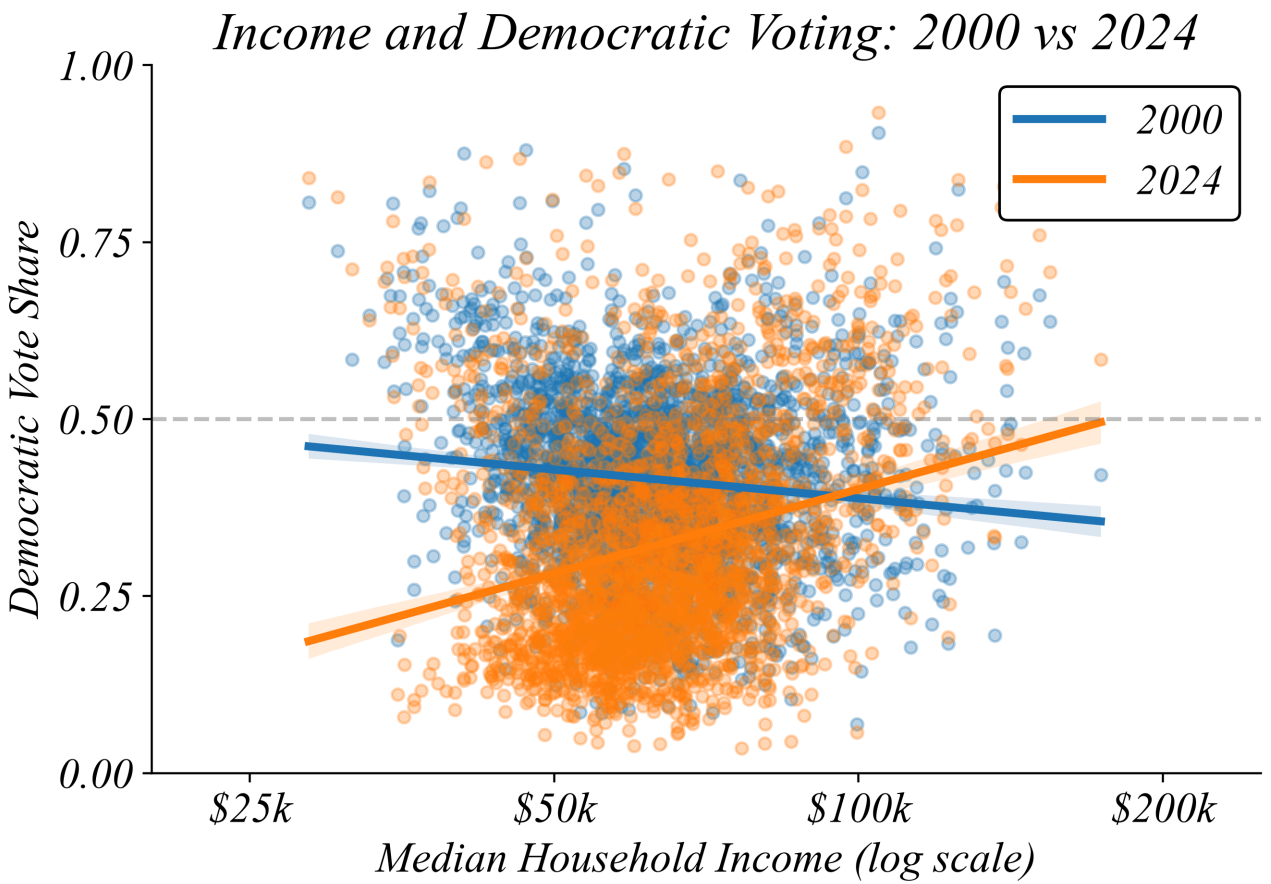
2 × 2 × 2 = 8

c. City A has log10(Population) = 4. City B has log10(Population) = 6. How many times larger is City B?

Two units apart on log10: 10 × 10 = 100 times larger

Reading Log Scales

On the income axis in Part 2.3, what does equal spacing mean?



*$25k → $50k → $100k → $200k: each step to the right is a **doubling** of income.*

Reading Log Scales

Why not just log every skewed variable?

In Part 2.2, most Starbucks rows had zero revenue.

- *log(0) is undefined, so logging a column of zeros fails.*
- *The zeros were a different kind of row: offers viewed, not purchases.*
- ***Filter first, then transform.***

Investigate Before You Visualize

Investigate: App Sessions

A coffee app records every session, whether or not the customer buys.

Session	Event	Minutes	Amount
S01	Purchased	4	6
S02	Browsed	2	0
S03	Purchased	9	11
S04	Browsed	7	0
S05	Purchased	15	17
S06	Browsed	12	0
S07	Purchased	6	8
S08	Browsed	3	0
S09	Purchased	12	14
S10	Browsed	10	0
S11	Purchased	18	21
S12	Browsed	5	0

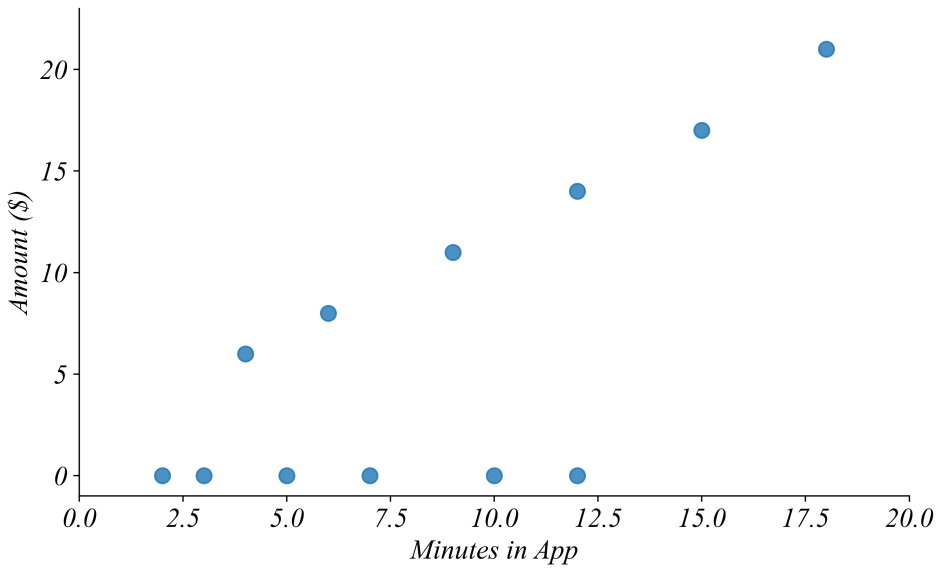
a. *Sketch a scatter of Minutes vs Amount using **all** the data.*

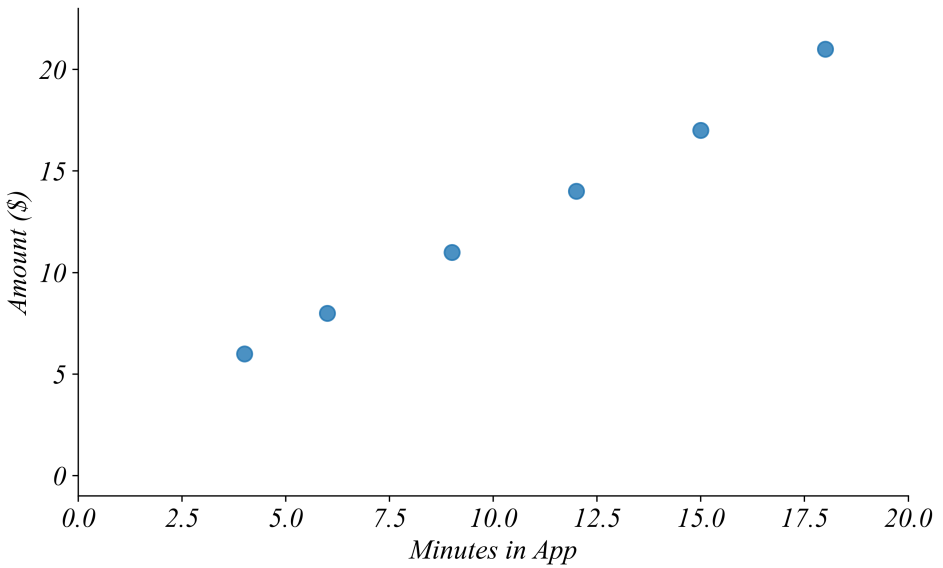
b. *Which rows should we drop to ask "do longer sessions lead to bigger orders?"*

c. *Sketch the scatter again after filtering.*

Investigate: App Sessions

All sessions vs. purchases only





All rows: half the points sit at zero and hide the pattern.

Filter `Event == 'Purchased'`*: longer sessions, bigger orders.*

Filter, Group, Summarize

Practice 1: Trace the Filter

Which apartments remain after filtering?

Apt	Area	Rent	Bedrooms
A1	Urban	1850	1
A2	Rural	900	3
A3	Suburban	1400	2
A4	Urban	1450	1
A5	Suburban	1650	3
A6	Rural	1200	2

Filter: `((Area == 'Urban') OR (Area == 'Suburban')) AND (Rent < 1500)`

Answer: A3, A4

Practice 2: Build the Filter

Construct the boolean logic from the pieces

Goal: Find all apartments that are Rural, OR that have at least 2 bedrooms AND rent under $1,500.

1. (Area == 'Rural')

2. (Bedrooms >= 2)

3. (Rent < 1500)

4. (Area == 'Urban')

Answer: 1 OR (2 AND 3)

Using the table from Practice 1: A2, A3, A6

Practice 3: Filter → Group → Summarize

The Part 2.2 pipeline, by hand

Txn	Offer	Event	Amount
T01	BOGO 5	Purchase	12
T02	BOGO 10	Viewed	0
T03	BOGO 5	Purchase	20
T04	Discount	Purchase	8
T05	BOGO 10	Purchase	30
T06	BOGO 5	Viewed	0
T07	Discount	Purchase	16
T08	BOGO 10	Purchase	18

Operations: 1) Filter for `Event == 'Purchase'` 2) Group by Offer 3) Calculate the mean and count of Amount

Practice 3: Filter → Group → Summarize

The Part 2.2 pipeline, by hand

Offer	Mean Amount	Count
BOGO 5	16	2
BOGO 10	24	2
Discount	12	2

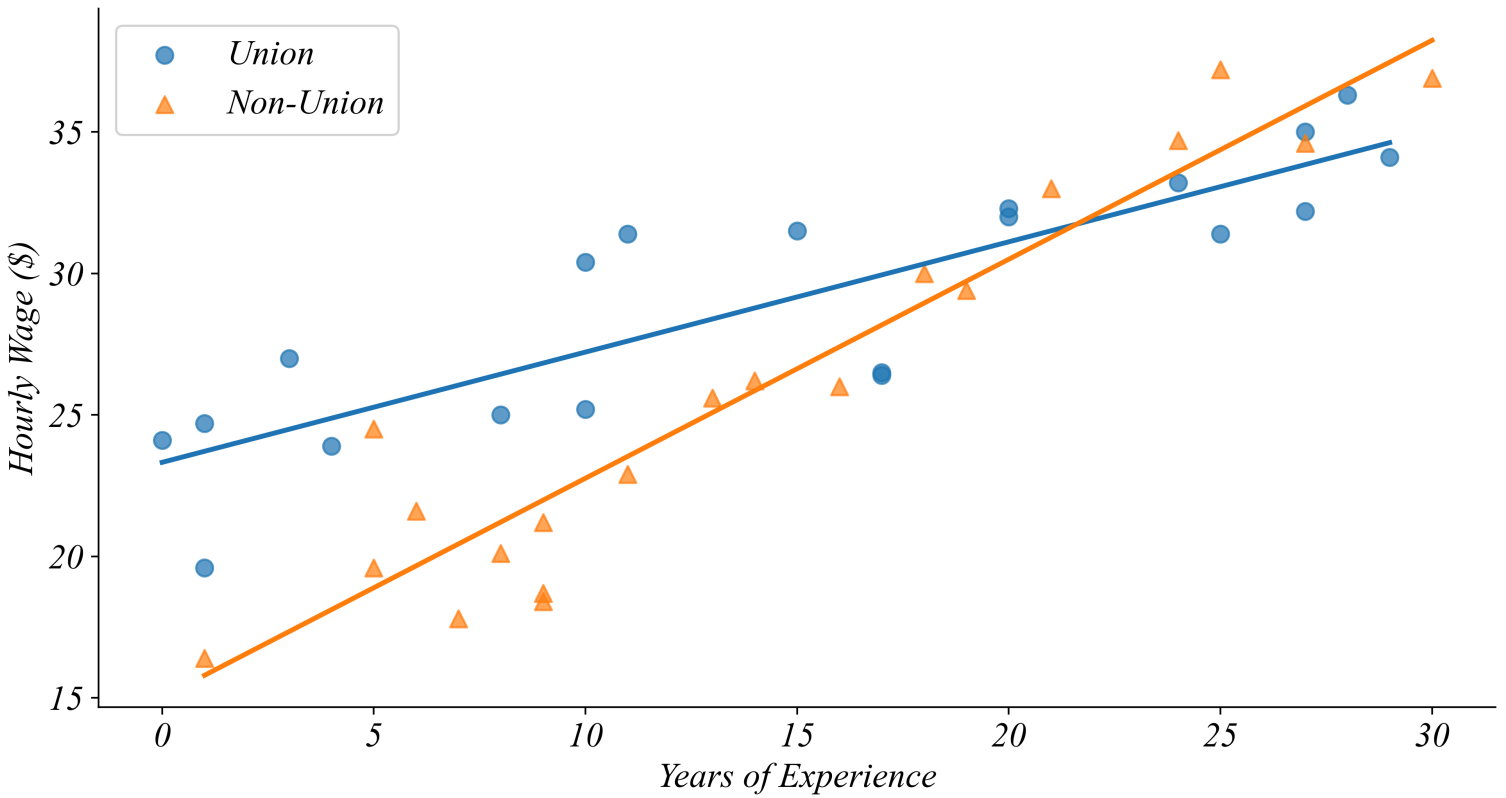
What if we forgot the filter? BOGO 5 becomes (12 + 20 + 0) / 3 = 10.67, and BOGO 10 becomes (0 + 30 + 18) / 3 = 16.

The viewed rows aren't purchases. Leaving them in drags every mean toward zero.

Relationships by Category

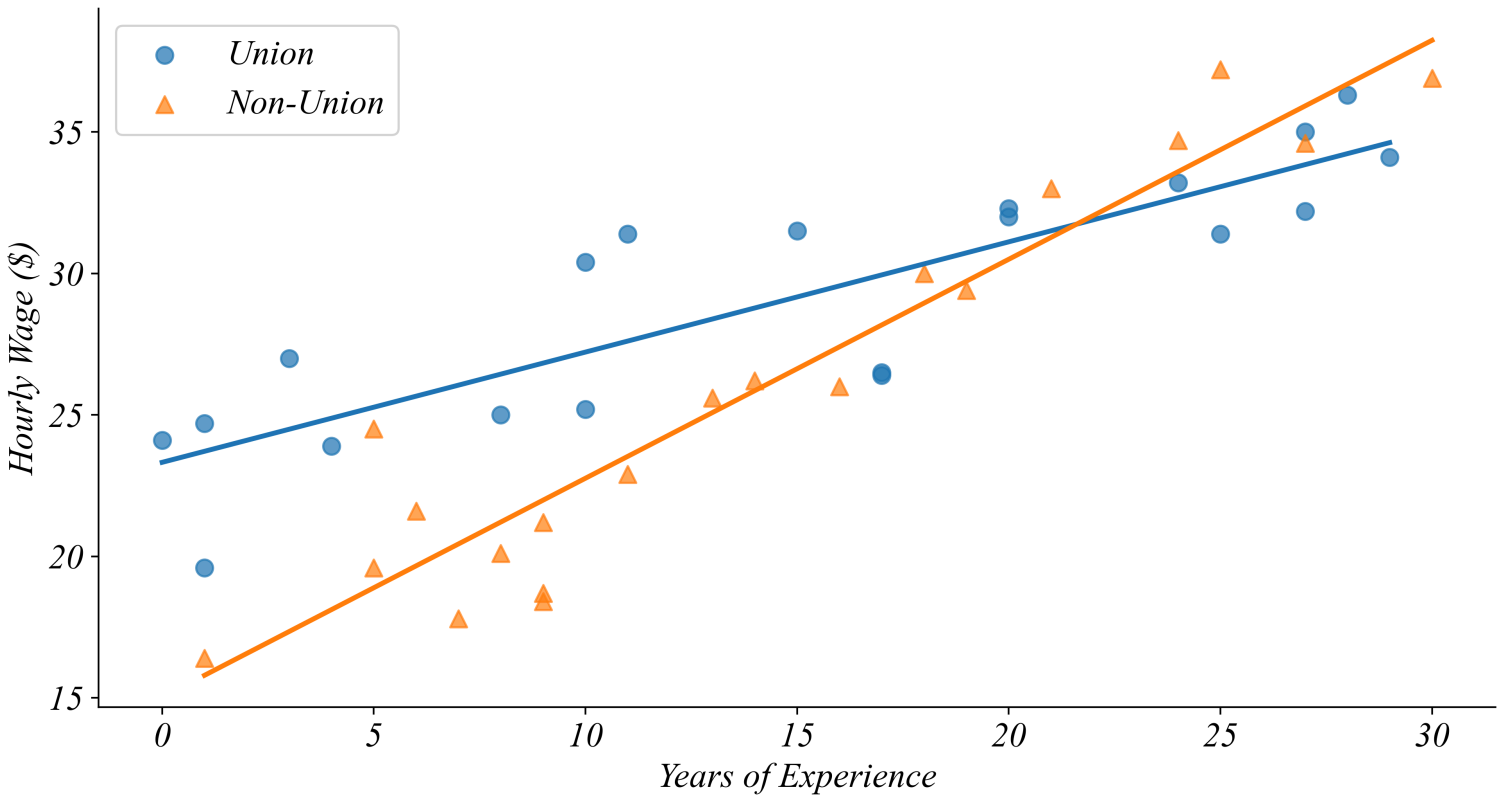
Read a Scatter by Category

Hourly wage and experience for union and non-union workers



Read a Scatter by Category

Hourly wage and experience for union and non-union workers

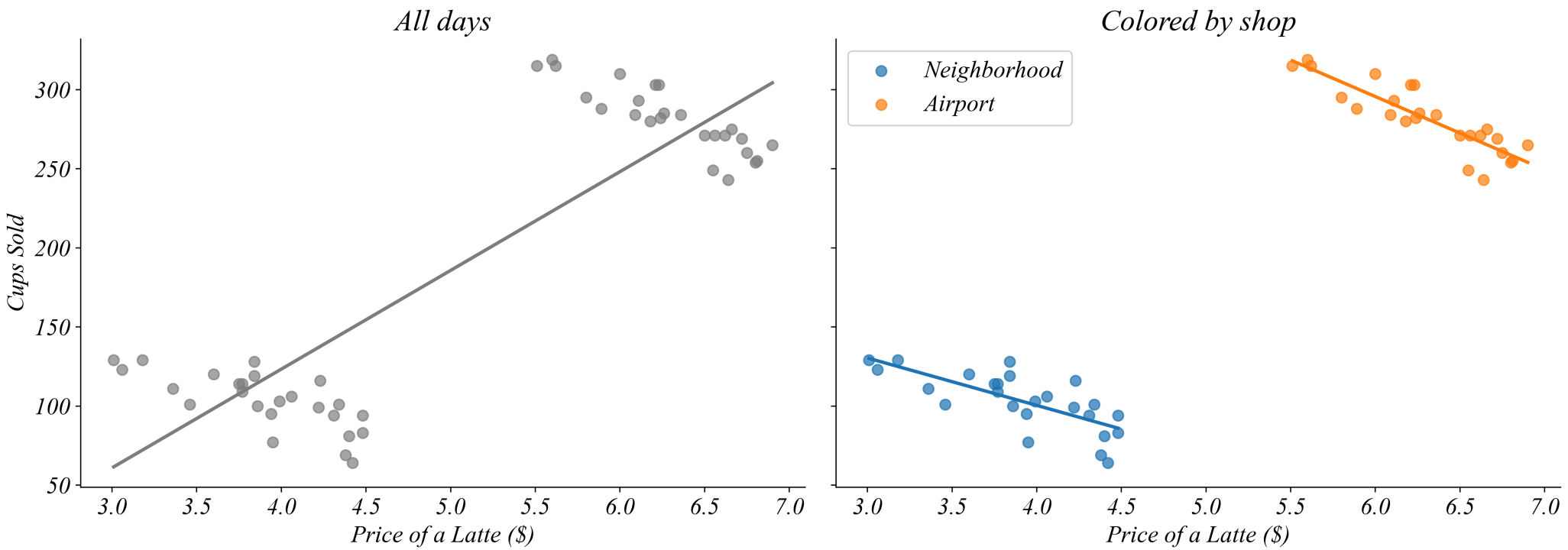


a. Both increase. b) Union at 5 years; non-union at 28 years.

b. Union workers start higher, but non-union wages rise faster with experience (a steeper line).

Why Color by Category?

Each point is one day at one coffee shop



All days together: higher prices, more cups sold. Is demand upward sloping?

*Within each shop, higher prices sell **fewer** cups. The airport just has higher prices **and** more customers.*

Why Color by Category?

The same lesson from Part 2.3

- *Counties shifted Republican on average (boxplot by year)*
- *Yet large, high-income counties vote Democratic (scatter)*
- *And the income relationship flipped between 2000 and 2024 (scatter by year)*
- ***A pattern in pooled data can differ from the pattern within each group.***

Combining and Reshaping Data

Practice 4: Trace the Merge

```
pd.merge(population, rents, on='City')
```

population

City	Population
Alder	120,000
Birch	850,000
Cedar	2,400,000
Dogwood	45,000

rents

City	Rent_2019	Rent_2024
Alder	950	1150
Birch	1300	1600
Cedar	2000	2300
Elm	800	980

How many rows in the merged data? Which cities drop out?

3 rows. Dogwood and Elm drop out: each appears in only one table.

Practice 5: Trace the Melt

Wide to long, so year can become a category

Wide

City	Rent_2019	Rent_2024
Alder	950	1150
Birch	1300	1600
Cedar	2000	2300

Long

City	Year	Rent
Alder	2019	950
Birch	2019	1300
Cedar	2019	2000
Alder	2024	1150
Birch	2024	1600
Cedar	2024	2300

3 rows become 6. Now `Year` is a column we can put on an axis or use for color.

Putting It Together

A Research Question

Do bigger cities have higher rents, and has that changed since 2019?

Using the Part 2.3 template, which figure answers each question?

1. ***Did rents rise?*** *Boxplot of Rent by Year*
2. ***Do bigger cities charge more?*** *Scatter of log Population vs Rent*
3. ***Has that relationship changed?*** *Scatter of log Population vs Rent, colored by Year*

Data steps: merge population and rents on City, melt the rent years, log population.

Part 2 | Summary

- ***Choose*** *the figure from the variable types: scatter, boxplot by category, or scatter by category*
- ***Filter*** *out rows that don't belong to the question before summarizing or plotting*
- ***Log*** *skewed variables: equal distances become equal multiples (but never log zeros)*
- ***Group*** *to summarize by category: one row per group*
- ***Merge*** *when the question needs variables from two sources;* ***melt*** *to turn columns into a category*